Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years building **MERN** and TypeScript applications with **Node.js**, **WebSockets** for real-time updates, MongoDB and SQL schemas, Docker, CI/CD on AWS, monitoring, and robotics motion-planning research.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized multi-tenant **database schemas** with row-level security and indexed queries for 500+ users, cutting dashboard latency from 1.2s to 350ms while keeping each tenant's data isolated.
- Built backend services and integration pipelines with validation and CI/CD-backed releases at 99%+ reliability, and restructured async job processing to triple **throughput** across 1K+ concurrent jobs without cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack **React** application with REST APIs and MySQL on AWS, deployed through GitHub Actions to Vercel and Render, serving 150+ instructors and 10K+ daily requests.
- Kept the system responsive at peak load with multithreaded batch processing, timeout handling, and fallback routing for 1K+ files per run, cutting instructor feedback turnaround 60% in **production**.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer for **autonomous** vehicle motion planning, with PyTorch pipelines over 100K+ trajectory logs, improving long-horizon trajectory planning 18% in closed-loop simulation across driving scenarios.
- Designed a **spatial** encoder fusing HD-map geometry with ego and nearby-agent trajectories in a shared coordinate frame, improving closed-loop planning stability 15% and reducing off-road trajectories in simulated driving.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **MERN** platform with TypeScript, Node.js and Express microservices, MongoDB, and **WebSocket** real-time updates for resource allocation and conflict resolution, syncing 2M+ records across multiple locations in real time.
- Automated Jenkins **CI/CD** with Jest, Selenium, and Snyk checks, Docker packaging, ARM-template infrastructure, and Sentry, Azure Monitor, and Kibana **monitoring**, cutting deployment time 50% across staged releases to production.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a React Native and React TypeScript app with Flask APIs and Firebase **real-time sync**, pushing live booking status to Android, iOS, and web clients and speeding confirmations 40%.
- Secured payment flows with JWT and wallet-signature **authentication** and idempotent webhooks for refunds and disputes, cutting manual handling about 70% across 50K+ booking and payment records for mobile and web users.

Skills

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS, Java
Frontend: React, Next.js, React Native, TailwindCSS, Responsive Design, Jest, Selenium, Git, Linux
Backend & Real-Time: Node.js, Express.js, WebSockets, REST APIs, Microservices, Firebase, JWT, Webhooks
Data: MongoDB, PostgreSQL, MySQL, Azure SQL, Schema Design, Query Optimization
DevOps & Cloud: Docker, Kubernetes, AWS, Azure, ARM Templates, CI/CD, Jenkins, GitHub Actions, Sentry, Kibana

Projects

Browser Robot Teleoperation Console | *React, TypeScript, WebRTC, Socket.IO, Node.js*

- Designed a **WebRTC** console that streams a robot's camera to the browser with low latency, sends joystick commands over **Socket.IO**, and shows live telemetry with heartbeat-based emergency stop.

3D Robot State Viewer | *React Three Fiber, Three.js, WebXR, Node.js, MongoDB*

- Designed a **React Three Fiber** viewer that renders a robot arm's joint state in 3D from streamed data, supports **WebXR** headsets for immersive inspection, and replays recorded sessions from MongoDB.

FinNexus Live | *Next.js, TypeScript, WebSockets, MongoDB, Kubernetes*

- Built a Next.js and TypeScript platform with **WebSocket** real-time updates, MongoDB storage, token-based secure access, and Kubernetes CI/CD, supporting 50,000+ users and 1M+ transactions with Plaid bank linking and filtering.